

$$\text{Dynamics: } m\ddot{x} + c\dot{x} + kx = u + d \quad (1)$$

$$\text{Control: } u = m[\ddot{x}_r + k_p(\dot{x}_r - \dot{x})] + c\dot{x} + kx + L \operatorname{sgn}(s) \quad (2)$$

$$\text{where } s = (\dot{x}_r - \dot{x}) + k_p(x_r - x) \quad (3)$$

Substitute (2) in (1)

$$\cancel{m\ddot{x}} + \cancel{c\dot{x}} + \cancel{kx} = m[\ddot{x}_r + k_p(\dot{x}_r - \dot{x})] + \cancel{c\dot{x}} + \cancel{kx} + L \operatorname{sgn}(s) + d$$
$$m[(\ddot{x}_r - \ddot{x}) + k_p(\dot{x}_r - \dot{x})] = -L \operatorname{sgn}(s) - d \quad (4)$$

$$\text{Derivative of (3) } \dot{s} = \ddot{x}_r - \ddot{x} + k_p(\dot{x}_r - \dot{x}) \quad (5)$$

Substituting (5) in (4)

$$\dot{s} = \frac{1}{m} [L \operatorname{sgn}(s) - d] \quad (6)$$

$$\text{Lyapunov function } V = 0.5 s^2$$

$$\dot{V} = s\dot{s} \quad (7)$$

Substituting ⑥ in ⑦

$$\dot{v} = s \left(\frac{1}{m} \right) (L \operatorname{sgn}(s) - d)$$

$$\dot{v} = -\frac{1}{m} (L|s| + ds) \quad (s \operatorname{sgn}(s) = |s|)$$

We will bound $L|s| + ds$ for $s > 0$ & $s < 0$

Case 1: $s > 0$ $L|s| + ds = (L + d)|s|$

$$\min (L + d)|s| = (L - D)|s| \quad \text{since } |d| < D$$

$$\Rightarrow s > 0 \quad L|s| + ds > (L - D)|s| \quad \textcircled{9}$$

Case 2: $s < 0$ Let $s = -t$

$$L|s| + ds = L|t| - d|t| = (L - d)|t|$$

Thus, $L|s| + ds = (L - d)|s|$

$$\min (L - d)|s| = (L - D)|s|$$

$$\Rightarrow s < 0 \quad L|s| + ds > (L - D)|s| \quad \textcircled{10}$$

From ⑨ and ⑩

$$L|s| + ds > (L-D)|s|$$

$$-(L|s| + ds) < -(L-D)|s| \quad \textcircled{11}$$

From ⑧ and ⑪

$$\dot{v} = -\frac{1}{m}(L|s| + ds) < -(L-D)|s|$$

$$\dot{v} < -(L-D)|s| \quad \text{Stability proof}$$

$$\boxed{L > D}$$

Condition on L